



Scenario 1: Pretrained Model Predicts Only One Class in Object Detection

Context: You fine-tuned a pretrained YOLOv5 model on a custom dataset (detecting helmets). After training, it always predicts only one class.

Problem: Despite having multiple classes in the dataset, the model outputs only “helmet” for every object

Answer:

- Check whether the dataset is balanced and whether all classes have enough training images.
- Verify that the class labels and YOLO configuration (nc and class names) are correct.
- Check the annotation files to make sure all classes are labeled correctly.

Scenario 2: Transfer Learning Model Trains Too Slowly on Small Dataset

Context: You are using a pretrained ResNet50 model for face mask classification (2 classes). You fine-tune the entire model on a small dataset (500 images).

Problem: Training takes too long and doesn't converge well.

Answer:

- Freeze most of the pretrained ResNet50 layers and train only the final classification layers.
- Also use a smaller learning rate and reduce the number of epochs if necessary.

Scenario 3: Transfer Learning Works on Training Data but Fails on New Environment

Context: You used a pretrained YOLOv8 model trained on outdoor images to detect tigers. But when used in night-mode CCTV footage, detection fails.

Problem: Domain shift between training data (bright daylight) and real-world application (night).

Answer:

- Add more night-time images to the training dataset.
 - Fine-tune the model using the new night-time data.
 - Use data augmentation such as brightness/contrast changes, noise, and low-light augmentation.
 - Validate on a separate night-time test/validation dataset.
-

Scenario 4: Object Detection Model Shows Bounding Boxes but Misses Object

Context: You trained a YOLOv5 model using transfer learning on a small vehicle dataset. Boxes are drawn, but not aligned with objects.

Problem: Bounding box coordinates are incorrect due to **wrong label format or image resizing** mismatch.

Answer:

- Check the label/annotation format - make sure bounding-box coordinates are in the correct YOLO format.
 - YOLO coordinates should be normalized between 0 and 1.
 - Ensure the image and bounding-box coordinates are transformed consistently.
 - Make sure the labels match the correct images.
 - Visualize the ground-truth bounding boxes before training to identify annotation problems.
 - Check the YOLO dataset configuration (data.yaml) and class IDs.
-

Scenario 5: Model Gives High mAP on Training But Low Real-Time Accuracy

Context: You used pretrained YOLOv8 for multi-class object detection (helmet, vest, mask). mAP on test set is 90%, but real-time inference on webcam is inaccurate.

Problem: Difference in resolution, camera angle, or image quality between webcam and training data. Poor generalization due to **dataset bias**.

Answer:

- Train with both high-quality and webcam-captured images.
- Check and correct bounding box annotations.

- Apply data augmentation (blur, brightness, rotation).
 - Increase dataset diversity by including different people, environments, and camera conditions.
 - Fine-tune YOLOv8 (epochs, learning rate, image size).
 - Create a validation set using webcam images instead of only relying on the original test set.
-

Scenario 6: VGG16 Model Very Slow During Training

Project: Fashion product classification using **VGG16**

Problem: Model takes too long to train even with a small dataset and batch size.

Answer:

- Use a smaller model (MobileNet, EfficientNet, ResNet18) instead of VGG16.
 - Reduce image size and batch size.
 - Use transfer learning and freeze VGG16 base layers.
 - Train on GPU instead of CPU.
 - Reduce epochs if training time is excessive.
-

Scenario 7: InceptionV3 Fails to Train Due to Input Size Mismatch

Project: Dog breed classification using **InceptionV3**

Problem: Runtime error due to input size mismatch.

Answer:

- Resize all images to 229 X 229 X 3.
 - Set the model input shape correctly (229 X 229 X 3).
 - Verify preprocessing and data generator settings.
 - Ensure image channels are RGB (3 channels).
-